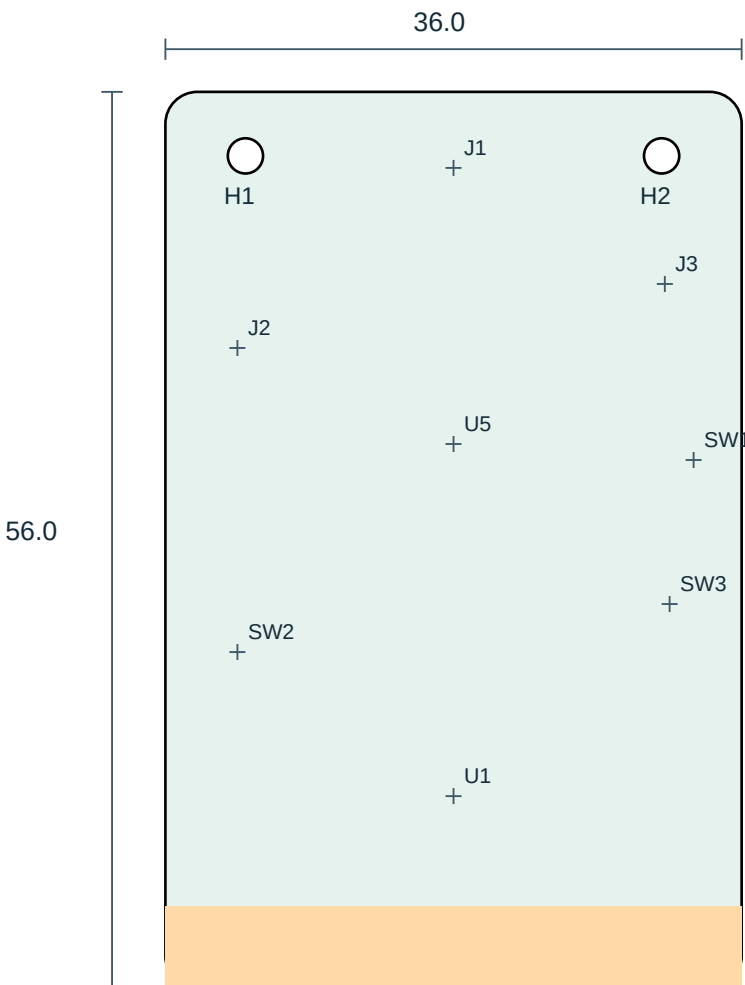


MAGNETIC SHUTTER REMOTE | ORIGINAL DESIGN R3

PCB outline, connector, actuator and antenna datums



36 x 56 x 1.0 FR4, 2 copper layers, R2 corners [A]

All 37 fitted electronics on TOP; bottom copper has no parts.

Board width changed to fit PH/SH connectors and top circuitry.

Outline +/-0.2; thickness +/-0.1 fabrication targets [B]

XY centre datum; +Y USB; +Z top face.

H1/H2: (+/-13,24), dia 2.2 NPTH [A]

Old harness opening W1 removed; wiring above board.

J1 C2894893: (0,23.25), rotation 180 deg [A,C]

Six contact USB-C: independent 5.1k CC1/CC2 pull-downs.

J2 PH (-13.5,12), mating from -Y, internal [A,D]

J3 SH (13.2,16), mating from +Z, lid removed [A,E]

SW1 (15,5), 90 deg; stroke 1.6 [A,F]

SW2 (-13.5,-7), SW3 (13.5,-4), top [A,G]

TMP390 thermal sensor (0,6), top [A,H]

Antenna exclusion X +/-18, Y=-28 to -22.9 [A,I]

No copper, wire, battery or metal hardware in strip.

[A] Original engineering: src/remote-circuit.tsx, dimensions.json.

[B] <https://jlcpcb.com/capabilities/pcb-capabilities>

[C] C2894893-manufacturer.pdf Rev A; no locating holes.

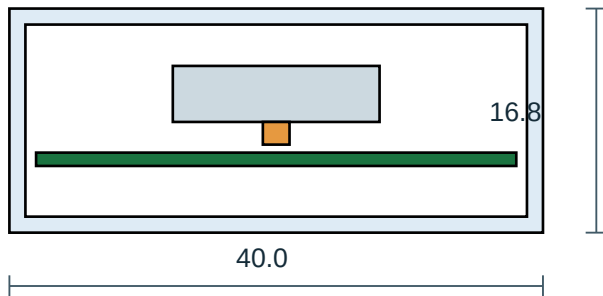
[D,E] JST-PH.pdf and JST-SH.pdf manufacturer land drawings.

[F,G] SHOUHAN-MSK12C02.pdf / C720477-manufacturer.pdf

[H,I] TI TMP390.pdf / Ebyte E73 manufacturer manual.

MAGNETIC SHUTTER REMOTE | ORIGINAL DESIGN R3

Remote enclosure, battery carrier and thermal stack



Z=0 external back; PCB bottom=5.0, top=6.0 [A]

Body 40 x 60 x 16.8; max slider width 42.4 [A]

Walls 1.2, inner cavity 37.6 x 57.6; corners R3 [A]

Lid underside 15.6; cap top 17.5 [A]

PCB-wall gap 0.8 nominal / 0.5 tolerance allocation.

Board support dia 4.2, Z=1.2..5; lid boss Z=6.15..15.6.

Magnet + 0.5 cap top 4.675; PCB gap 0.325 nominal.

DATA POWER DTP401525(PHR), external pack [J]

Maximum 15.5 x 27 x 4.2; side space 16.6 [A,J]

Pack X +/-7.75; Y=-5.7..21.3; Z=8.3..12.5 [A]

Carrier floor Z=7.9..8.3; side gaps 0.55 nominal [A]

Carrier thermal aperture 2.4 x 2.4 at (0,6).

TMP390 package maximum height 0.6 -> top Z=6.6 [H]

Thermal-pad assembled gap target 1.7 [A,K]

Actual chip height/contact require measurement + shims.

Pouch must not be compressed by lid, shims or pad.

C8 maximum height 1.45: top Z=7.45 [L]

C8 relief 5 x 3 at (5,17.5); battery gap 0.85 nominal.

Battery-to-lid clearance 3.1 nominal, hardware untested.

PH mated max height 5.5 -> top Z=11.5 [D]

SH mated max height 6.3 -> top Z=12.3 [E]

Both connectors outside battery envelope in X.

[J] DTP401525-PHR-v2.pdf manufacturer pack drawing.

[K] BERGQUIST GP1500-0.080-02-0404, free 2.032.

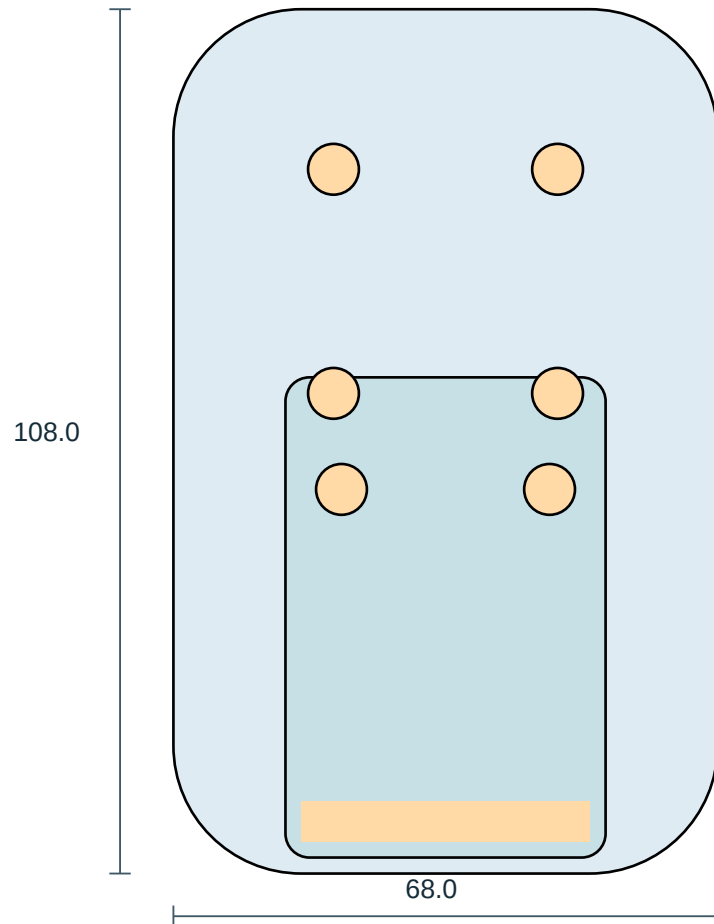
Henkel thermal-pad datasheet; cut 2 x 2, nominal 16.3% strain.

[L] Samsung CL21A106KAYNNNE package drawing.

No JJC internal dimensions are asserted.

MAGNETIC SHUTTER REMOTE | ORIGINAL DESIGN R3

Grip, docking envelope, magnets and antenna margins



Grip 68 x 108, base 6.5; rail top Z=9.8; R16 [A]

Dock centre Y=-42; back Z=6.8; button top 24.3 [A]

Plate Z=-0.5; total plate-to-button stack 24.8 [A]

Cradle inner width 40.6; nominal side gap 0.3 [A]

Rails 1.6 x 48 x 3.3, centres X +/-21.1, Y=-38.

Stop 40.6 x 1.6 x 1.2, centre Y=-10.9.

Eight K&J D42 magnets, dia 6.35 x 3.175 [M]

Pockets dia 6.65; nominal adhesive radial gap 0.15 [A]

Remote (+/-13,16); dock (+/-13,-26) [A]

Phone magnets (+/-14,+/-14), captive behind cover [A]

Separate steel phone plate 45 x 45 x 0.5 [A]

Phone/case keep-in envelope 84 x 180 x 10 [A]

Phone bottom aligned Y=-35 +/-5; mark alignment [A]

Antenna Y=-70..-64.9 in dock frame.

Nearest permitted phone metal Y=-40: 24.9 gap.

Detached battery planar gap to antenna strip: 17.2.

Chosen metal margin 15; Ebyte provides no numeric rule.

These are calculated margins; RF hardware tests pending.

[M] <https://www.kjmagnetics.com/d42-neodymium-disc-magnet>

MAGNETIC SHUTTER REMOTE | ORIGINAL DESIGN R3

USB access, actuator travel, wiring and acceptance schedule

USB-C HCTL HC-TYPE-C-6P-01A C2894893; Rev A body: 8.95 x 6.85 x 3.16 +/-0.15 height [C]

Recommended slots 4 x 0.50 x 1.40; shell centres X +/-4.32, row pitch 3.70 [C].

Supplier lands: GND 0.80, VBUS 0.76, CC 0.70 wide x 1.20; shell copper 1.10 x 1.80.

Opening 12 x 7 at (0,29,8), through front wall; connector mates +Y after 180 deg rotation [A,C].

Both USB orientations use separate CC pull-downs; no USB data or USB bootloader.

Plug reservation max 10 x 6.5 is an enclosure allocation; test intended cables before enclosure release.

Shutter/pair caps dia 5.8 in dia 6.2 holes; flange dia 8 x 0.8; stem dia 1.6 x 6.6 [A].

Stem rest Z=8.2; switch nominal top 8.0; free gap 0.2; travel 0.2 +/-0.1 [A,G].

TPU stem compliance is intentional; adjust to actual assembled height; avoid preload.

Side slider fork width 1.7 around nominal 1.3 switch actuator; designed travel 1.6 [A,F].

Harness: pack supplier lead length 100 +/-3 mm, red BAT+, black GND; actual connector polarity must be verified [J].

Two insulated conductor reservations dia 1.2, centre spacing 1.3; loop above left PH bay and outside antenna strip [A].

Harness bends and slack are assembly reservations; validate actual bend radius ≥ 3 , abrasion and strain relief.

No wire-through PCB opening or underside electronic assembly remains.

Before release: manufacturing copper/holes/slots, full CAM read, strict CPL rotations, stencil + hybrid shell process.

The manufacturer pattern does not automatically establish process compliance.

Physical acceptance still pending: battery protection, charge current/voltage, temperature cutoff and thermal coupling;

pouch clearance, wire retention, button/slider travel, magnet retention/drop tests, RF docked and detached;

firmware programming and native iPhone Camera / representative Android camera pairing and shutter behavior.

All electronics on TOP does not require single-layer copper. Two-layer FR4 remains selected.

This drawing is an original mechanical design. Calculated fit is not a physical measurement or acceptance test.